

## **Problem Statement – 17**

**Title of Problem Statement:** Single-Pass Drone Video to Accurate 3D Model Generation System

**Background:** Generation of accurate 3D models of buildings, infrastructure, terrain, and objects typically requires multiple drone passes, extensive image overlap, specialized flight planning, and significant post-processing time. In operational scenarios such as disaster response, surveillance, infrastructure inspection, military reconnaissance, and rapid mapping, there is often only a single opportunity to capture data over the target area. A solution capable of generating an accurate and textured 3D model from a single drone pass video would significantly reduce mission time, operator effort, data acquisition requirements, and processing complexity while enabling near real-time situational awareness.

**Description:** Design and develop an AI-enabled system capable of generating a georeferenced and metrically accurate 3D model of a scene using only a single-pass drone video stream captured from a moving UAV. The system should process video frames captured during one flight path and reconstruct:

- (i) 3D terrain and structures
- (ii) Building facades and rooftops
- (iii) Roads and infrastructure
- (iv) Vegetation and obstacles
- (v) Textured 3D meshes or point clouds

### **Expected Solution/Deliverables:**

The generated model should be suitable for visualization, measurement, and analysis purposes.

### **Key Challenges**

- (i) Limited viewing angles due to single flight path.
- (ii) Motion blur and video compression artifacts.
- (iii) Variable illumination and shadows.
- (iv) Dynamic objects (vehicles, humans, animals).
- (v) GPS inaccuracies and sensor noise.
- (vi) Real-time or near-real-time processing requirements.

- (vii) Reconstruction of occluded surfaces.
- (viii) Maintaining metric accuracy without extensive Ground Control Points (GCPs).

### **Input Data :**

#### **Mandatory**

- (i) Drone video (1080p/4K)
- (ii) GPS coordinates
- (iii) Flight metadata

#### **Optional**

- (i) IMU data
- (ii) Barometric altitude
- (iii) Camera intrinsic parameters
- (iv) RTK/PPK corrections

### **Desired Output:**

| Parameter           | Target                                   |
|---------------------|------------------------------------------|
| Reconstruction Type | 3D Mesh / Point Cloud                    |
| Processing Time     | < 15 minutes for 10-minute video         |
| Spatial Accuracy    | $\leq 1$ m                               |
| Coverage            | Entire visible scene                     |
| Output Formats      | OBJ, PLY, LAS, GeoTIFF, .glb/.gltf, .fbx |
| Visualization       | Web-based or Desktop Viewer              |
|                     |                                          |

### **Evaluation Criteria:**

| Criteria                | Weightage |
|-------------------------|-----------|
| Reconstruction Accuracy | 30%       |
| Model Completeness      | 20%       |
| Processing Speed        | 20%       |
| Innovation              | 15%       |
| Scalability             | 10%       |
| User Interface          | 5%        |

### **Potential Applications :**

- (i) Border and strategic area mapping
- (ii) Disaster damage assessment
- (iii) Urban planning and smart cities
- (iv) Infrastructure inspection
- (v) Construction progress monitoring
- (vi) Archaeological documentation
- (vii) Digital twin generation
- (viii) Military reconnaissance and mission planning

**Organisation:** National Technical Research Organisation

**Category:** Software

**Theme:** Drone/ Robotics

**Youtube Link/Video Link:** Nil

**Dataset Link:** Will be provided real time.